Junseo Park

Curriculum Vitae

Integrated M.S.–Ph.D. student · High-Impact Weather Prediction Laboratory

Department of Civil, Urban, Earth, and Environmental Engineering, UNIST

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Education

Integrated M.S.–Ph.D. program, UNIST

Department of Civil, Urban, Earth, and Environmental Engineering · 27 Feb 2023–present

B.S. in Urban and Environmental Engineering, UNIST

1 Mar 2020–16 Feb 2023

Research Interests

Regional climate modeling and CORDEX, atmosphere–ocean coupled modeling, and AI-based climate emulators.

Journal Publications

Kim, Y.-H., Cho, W., Kang, T., **Park, J.**, Park, C., Kim, T., Cha, D.-H., Jun, S.-Y., Min, S.-K. (2026). Assessing anthropogenic thermodynamic influences on marine heatwaves near the Korean Peninsula using a coupled regional climate model. *Frontiers in Marine Science*, 13, 1932600. Published 12 Nov 2026. 10.3389/fmars.2026.1932600

Mun, T., Park, H., **Park, J.**, Cha, D.-H., Lee, M.-I., Son, S.-W., Min, S.-K. (2025). Spatiotemporal Characteristics of Mesoscale Convective Systems Over East Asian Monsoon Region Simulated by a Convection-Permitting Model. *Journal of Geophysical Research: Atmospheres*, 130(24), e2025JD044820. 10.1029/2025JD044820

Park, J. and Cha, D.-H. (2025). Assessment of Planetary Boundary Layer and Cumulus Parameterization Schemes on 2022 East Asian Summer Monsoon Simulation. *Journal of Climate Research*, 20(1), 27–42. 10.14383/cri.2025.20.1.27

Awards and Scholarships

Best Poster Paper Award

Korean Society of Climate Change Research, Fall Meeting · Nov 2024. Awarded for Differences in WRF Simulations of 2022 East Asian Summer Precipitation across Planetary Boundary Layer and Cumulus Parameterization Schemes, with Taeho Mun and Dong-Hyun Cha. Award ceremony program

Encouragement Award, Disaster and Safety Data Hackathon

Director's Award, UNIST Industry–University Cooperation Foundation · 13 Jun 2023. Team members Woojin Cho, Haeun Jo, Taehun Kang, and Junseo Park. Event announcement

UNIST Scholarship (Graduate)

Ulsan National Institute of Science and Technology · 27 Feb 2023-present

UNIST Scholarship (Undergraduate)

Ulsan National Institute of Science and Technology · 1 Mar 2020-16 Feb 2023

Conference Presentations

2026

River Discharge and Ocean Spin-Up Effects on Coupled Regional Climate Simulations Over CORDEX East Asia
AOGS Annual Meeting · Oral (AS102) · Fukuoka, Japan · 4 Aug 2026
Junseo Park, Woojin Cho, Seok-Woo Shin, Dong-Hyun Cha

Role of Ocean Initialization and River Discharge in Coupled Regional Climate Simulations over East Asia
KSCC First-Half Meeting · Poster · Seoul, South Korea · 25 Jun 2026
Junseo Park, Woojin Cho, Seok-Woo Shin, Uijeong Han, Dong-Hyun Cha

Effects of River Discharge and Ocean Initialization on Coupled Regional Climate Simulations over East Asia
Korean Meteorological Society Spring Meeting · Poster · Busan, South Korea · 11 May 2026
Junseo Park, Woojin Cho, Seok-Woo Shin, Dong-Hyun Cha

Ocean Spin-Up Effects on Coupled Regional Climate Simulations over CORDEX East Asia
CORDEX East Asia Workshop · Poster · Kyoto, Japan · 5 Mar 2026 Program
Junseo Park, Woojin Cho, Seok-Woo Shin, Dong-Hyun Cha

2025

Impact of Atmosphere-Ocean Coupling on Regional Climate Downscaling over East Asia
AOGS Annual Meeting · Oral (AS49) · Singapore · 29 Jul 2025
Junseo Park, Woojin Cho, Seok-Woo Shin, Taehyung Kim, Dong-Hyun Cha

Performance Evaluation of a Regional Climate Model with Atmosphere–Ocean Interactions
Korean Meteorological Society Spring Meeting · Poster · Seoul, South Korea · 13 May 2025 Abstract
Junseo Park, Woojin Cho, Seok-Woo Shin, Taehyung Kim, Dong-Hyun Cha

Influence of PBL-CPS Combinations and Resolution on the Simulation of East Asian Summer Monsoon Precipitation in 2022 Using WRF
EGU General Assembly · Poster (EGU25-8015) · Vienna, Austria · 28 Apr 2025 Abstract
Junseo Park, Dong-Hyun Cha, Dong-Kyou Lee

2024

Effects of Planetary Boundary Layer and Cumulus Parameterization Schemes on the Simulation of the 2022 East Asian Summer Monsoon
Korean Meteorological Society Fall Meeting · Poster (P-225) · Daegu, South Korea · 29 Oct 2024 Abstract
Junseo Park, Taeho Mun, Dong-Hyun Cha

Analysis of Differences in WRF Simulations of East Asian Summer Monsoon in 2022 Based on Planetary Boundary Layer and Convective Parameterization Schemes
International Conference on Mesoscale Convective Systems XVI · Poster (P-016) · Gyeongju, South Korea · 14 Oct 2024 Program
Junseo Park, Taeho Mun, Dong-Hyun Cha

Sensitivity of East Asian Summer Precipitation in 2022 to Boundary Layer and Convection Schemes
AOGS Annual Meeting · Poster (AS65) · Pyeongchang, South Korea · 25 Jun 2024
Junseo Park, Eunji Kim, Haeun Jo, Taeho Mun, Dong-Hyun Cha

Relationship Between Arctic Oscillation and PM2.5 Concentrations in Seoul: an Analysis Using ERA5 Data and WRF Model Assessments

AOGS Annual Meeting · Poster (AS75) · Pyeongchang, South Korea · 25 Jun 2024
Junseo Park, Woojin Cho, Taeho Mun, Hongjun Choi, Dong-Hyun Cha, Sung-Chul Hong, Min-Hyeok Choi, Jae-Bum Lee

Differences in WRF Simulations of 2022 East Asian Summer Precipitation across Planetary Boundary Layer and Cumulus Parameterization Schemes

KSCC First-Half Meeting · Poster (P-13) · Seoul, South Korea · 20 Jun 2024 Program
Junseo Park, Taeho Mun, Dong-Hyun Cha

2023

Differences in WRF Simulations of 2022 East Asian Summer Precipitation across Planetary Boundary Layer Parameterization Schemes

KSCC Fall Meeting · Poster (P-14) · Gyeongju, South Korea · 23 Nov 2023 Program
Junseo Park, Taeho Mun, Dong-Hyun Cha

Impact of increasing model resolution on added values in regional climate simulation of heavy precipitation

International Conference on Regional Climate–CORDEX · Poster (A2-P-21) · 26 Sep 2023 Abstract
Junseo Park, Gayoung Kim, Jineun Kim, Dong-Hyun Cha

Co-authored Conference Contributions

Co-authorship and presenters are recorded in the laboratory conference list.

2026

Role of Air–Sea Interaction in Modulating Winter Temperature Biases over Northeast Asia in a Coupled Regional Climate Model

AOGS Annual Meeting · Poster · Fukuoka, Japan · 5 Aug 2026
Role · Co-author | Presenter · Seok-Woo Shin
Contributors · Seok-Woo Shin, Minkyu Lee, Changyong Park, Woojin Cho, **Junseo Park**, Dong-Hyun Cha
Record

Development of Downscaled Climate Change Scenarios Using a Coupled Regional Climate Model

KSCC First-Half Meeting · Oral · Seoul, South Korea · 25 Jun 2026
Role · Co-author | Presenter · Dong-Hyun Cha
Contributors · Dong-Hyun Cha, **Junseo Park**, Woojin Cho, Seok-Woo Shin, Uijeong Han Record

Mechanisms Improving Northeast Asian Winter Temperature Simulations in an Atmosphere–Ocean Coupled Regional Climate Model

KSCC First-Half Meeting · Oral · Seoul, South Korea · 25 Jun 2026
Role · Co-author | Presenter · Seok-Woo Shin
Contributors · Seok-Woo Shin, Minkyu Lee, Changyong Park, Woojin Cho, **Junseo Park**, Uijeong Han, Dong-Hyun Cha Record

Establishing a High-Resolution Dynamical Downscaling Framework for East Asia Based on CORDEX-CORE

KSCC First-Half Meeting · Poster · Seoul, South Korea · 25 Jun 2026
Role · Co-author | Presenter · Uijeong Han
Contributors · Uijeong Han, Seok-Woo Shin, Taeho Mun, **Junseo Park**, Dong-Hyun Cha Record

Winter Temperature Biases over Northeast Asia: An Atmosphere–Ocean Coupled Regional Climate Modeling Analysis

Korean Meteorological Society Spring Meeting · Poster · Busan, South Korea · 11 May 2026
Role · Co-author | Presenter · Seok-Woo Shin
Contributors · Seok-Woo Shin, Minkyu Lee, Changyong Park, Woojin Cho, **Junseo Park**, Uijeong Han, Dong-Hyun Cha Record

Understanding Winter Temperature Biases over Northeast Asia: Insights from Air–Sea Coupled Regional Climate Modeling
CORDEX East Asia Workshop · Poster · Kyoto, Japan · 5 Mar 2026
Role · Co-author | Presenter · Seok-Woo Shin
Contributors · Seok-Woo Shin, Minkyu Lee, Changyong Park, Woojin Cho, **Junseo Park**, Dong-Hyun Cha
Record

2025

Simulating Western North Pacific Tropical Cyclones Using a Coupled Regional Climate Model
KSCC Fall Meeting · Oral · Busan, South Korea · 3-5 Dec 2025 (conference dates)
Role · Co-author | Presenter · Dong-Hyun Cha
Contributors · Dong-Hyun Cha, **Junseo Park**, Woojin Cho, Seok-Woo Shin Record

Effects of the East Asian Winter Monsoon on Snowfall over the Korean Peninsula: A Regional Climate Model Approach
KSCC Fall Meeting · Oral · Busan, South Korea · 3-5 Dec 2025 (conference dates)
Role · Co-author | Presenter · Seok-Woo Shin
Contributors · Seok-Woo Shin, Minkyu Lee, Taeho Mun, **Junseo Park**, Jaeyeong Kim, Dong-Hyun Cha Record

Impact of Air-Sea Interaction on the Winter Temperature Simulation over East Asia with a Regional Climate Model
AOGS Annual Meeting · Oral · Singapore · 29 Jul 2025
Role · Co-author | Presenter · Seok-Woo Shin
Contributors · Seok-Woo Shin, Minkyu Lee, Woojin Cho, **Junseo Park**, Changyong Park, Dong-Hyun Cha
Record

Running a Coupled Regional Climate Model over East Asia with Ocean Initialization
KSCC First-Half Meeting · Oral · Seoul, South Korea · 19 Jun 2025
Role · Co-author | Presenter · Dong-Hyun Cha
Contributors · Dong-Hyun Cha, **Junseo Park**, Woojin Cho, Seok-Woo Shin Record

Winter Temperature Biases over East Asia in a Regional Climate Model: Atmosphere–Ocean Coupling and PBL Parameterization Schemes
KSCC First-Half Meeting · Oral · Seoul, South Korea · 19 Jun 2025
Role · Co-author | Presenter · Seok-Woo Shin
Contributors · Seok-Woo Shin, Changyong Park, Minkyu Lee, Woojin Cho, **Junseo Park**, Eunji Kim, Uijeong Han, Dong-Hyun Cha Record

Effects of Atmosphere–Ocean Interactions on Regional Climate Model Simulations of East Asian Winter Temperatures
Korean Meteorological Society Spring Meeting · Oral · Seoul, South Korea · 13 May 2025
Role · Co-author | Presenter · Seok-Woo Shin
Contributors · Seok-Woo Shin, Minkyu Lee, Woojin Cho, **Junseo Park**, Changyong Park, Dong-Hyun Cha
Record

Evaluating Particulate Matter Forecasts over Seoul by Arctic Oscillation Phase
Korean Meteorological Society Spring Meeting · Oral · Seoul, South Korea · 12 May 2025
Role · Co-author | Presenter · Taehyung Kim
Contributors · Taehyung Kim, **Junseo Park**, Mihye Seo, Dong-Hyun Cha, Sung-Chul Hong, Jae-Bum Lee Record

2024

Regional Climate Simulation over East Asia Using a Coupled Regional Climate Model
KSCC Fall Meeting · Oral · Jeju, South Korea · 28 Nov 2024
Role · Co-author | Presenter · Dong-Hyun Cha
Contributors · Dong-Hyun Cha, **Junseo Park**, Seok-Woo Shin Record

2023

Recent Changes and Related Factors of Tropical Night in the South Korea Metropolitan Area
AOGS Annual Meeting · Poster · 1 Aug 2023
Role · Co-author | Presenter · Taehun Kang
Contributors · Taehun Kang, Donghyuck Yoon, **Junseo Park**, Dong-Hyun Cha Record

Synoptic Patterns and Sea-Surface Temperature Effects on Tropical Nights in the Seoul Metropolitan Area

Korean Meteorological Society Spring Meeting · Poster · 20 Apr 2023
Role · Co-author | Presenter · Taehun Kang
Contributors · Taehun Kang, Donghyuck Yoon, **Junseo Park**, Dong-Hyun Cha Record

Research Projects

Periods shown are those recorded in the supplied research record.

Diagnosing the Causes of Subtropicalization of Precipitation over the Korean Peninsula Using the Multifaceted Capabilities of Convection-Permitting Models

1 May 2024-30 Apr 2027 · Participating Researcher
Lead institution · Ulsan National Institute of Science and Technology
Grant number · RS-2024-00350141

Development of Technologies for Diversifying Regional Climate Change Information

1 Apr 2024-31 Dec 2026 · Participating Researcher
Lead institution · Pohang University of Science and Technology
Grant number · RS-2024-00403386

Development of Technologies to Improve Intraseasonal Physical and Dynamical Processes in a Climate Prediction System

1 Mar 2024-31 Dec 2026 · Participating Researcher
Lead institution · Seoul National University

Dual-Polarization Radar Data Assimilation Using a Local Ensemble Kalman Filter

1 Oct 2023-31 Dec 2023 · Participating Researcher
Lead institution · Kyungpook National University

Effects of Multiscale Physical Parameterization Schemes on Mesoscale Simulations Using a High-Resolution Numerical Model

1 Sep 2023-29 Feb 2024 · Participating Researcher
Lead institution · Ulsan National Institute of Science and Technology
Grant number · NRF-2022R1F1A1076381

Investigation and prediction system development of marine heatwave around the Korean Peninsula originated from the subarctic and western Pacific

1 Jun 2023-31 Aug 2023 · Participating Researcher
Lead institution · Korea Institute of Ocean Science and Technology
Grant number · 20190344 / RS-2019-KS191260

Development of Technologies for Producing and Analyzing Disaster Prevention Information Based on AR6 SSP-RCP Scenarios

1 Jan 2023-31 Dec 2023 · Participating Researcher
Lead institution · Ulsan National Institute of Science and Technology
Grant number · KMI2021-00913